

1. Problem Statement

Traditional networks lack a unified view of flow rules across multiple switches. Administrators must log into each device individually to inspect tables, identify stale rules, or track traffic. This project implements a Multi-Switch Flow Table Analyzer for an SDN environment, a Python tool that reads OVS flow tables directly using `ovs-ofctl`, displaying rule details, classifying rules as ACTIVE/NEW/UNUSED, and supporting live-refresh monitoring.

2. SDN Architecture

The system uses Mininet with a POX Remote Controller as the control plane and Open vSwitch as the data plane. Three switches (s1, s2, s3) are connected in a full mesh with six hosts (two per switch). The analyzer queries tables via `ovs-ofctl` commands, making it independent of controller internals.

Switch	Hosts	Host IPs	Inter-switch Links
s1	h1, h2	10.0.0.1, 10.0.0.2	s2, s3
s2	h3, h4	10.0.0.3, 10.0.0.4	s1, s3
s3	h5, h6	10.0.0.5, 10.0.0.6	s1, s2

Table 1: Network Topology

3. Controller & Flow Rule Management

The POX controller (`controller.py`) installs a priority-0 table-miss rule on each switch at startup, forwarding unmatched packets to the controller. On `PacketIn`, it learns the source MAC/port mapping and installs a priority-1 unicast forwarding rule (idle timeout: 30s, hard timeout: 120s). The analyzer classifies rules as:

- **ACTIVE** — `n_packets > 0` (matched traffic)
- **NEW** — `n_packets == 0`, `duration < 30s` (recently installed)
- **UNUSED** — `n_packets == 0`, `duration ≥ 30s` (stale)
- **DEFAULT** — `priority == 0` (table-miss, sends to controller)

Rule efficiency = (Active Rules / Total Rules) × 100%

4. Implementation

Three scripts implement the system. `topology.py` creates the Mininet network and seeds initial traffic using a POX RemoteController. `controller.py` runs as a POX component, handling MAC learning and flow installation. `analyzer.py` provides three modes: (1) Single snapshot: formatted table of all flow entries per switch with aggregate summary; (2) Live refresh: continuous polling with terminal refresh every 5 seconds; (3) Raw dump: unprocessed `ovs-ofctl` output for debugging.

(PYTHON 3.10 SCRIPTS)

GITHUB LINK: <https://github.com/Chiranth-D-Nandi/cn-individual-assignment/>

5. Performance Evaluation

```
*** Cleanup complete.
*** Adding controller
*** Adding 3 switches
*** Adding 6 hosts
*** Adding links
*** Starting network
*** Configuring hosts
h1 h2 h3 h4 h5 h6
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Waiting for network to settle...
*** Generating initial traffic...

*** Network ready! Switches: s1, s2, s3
*** Hosts: h1(10.0.0.1) h2(10.0.0.2) h3(10.0.0.3)
***          h4(10.0.0.4) h5(10.0.0.5) h6(10.0.0.6)
*** Run 'pingall' to generate traffic
*** Open another terminal and run: sudo python3 analyzer.py

*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> X X X X X
h2 -> X X X X X
h3 -> X X X X X
h4 -> X X X X X
h5 -> X X X X X
h6 -> X X X X X
*** Results: 100% dropped (0/42 received)
```

Figure 1: Mininet startup + pingall (100% drop expected)

Analyzer Menu

```
vboxusercdn@UbuntuVM:~/Documents/table_analyser$ sudo python3 analyzer.py

=====
MULTI-SWITCH FLOW TABLE ANALYZER
=====

No Ryu needed – reads OVS tables directly

1 Single analysis snapshot
2 Dynamic monitor (live refresh)
3 Raw dump (debug / see everything)
q Quit

Choice: █
```

Figure 2: Analyzer main menu

Mode 1 — Snapshot

COMPUTER NETWORKS INDIVIDUAL ASSIGNMENT SDN MININET**SUBJECT CODE: UE24CS252****STUDENT NAME: CHIRANTH D NANDI****SRN: PES2UG24AM048**

Choice: 1

=====

MULTI-SWITCH FLOW TABLE ANALYZER

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Time: 2026-04-16 18:22:16

Switches found: ['s1', 's2', 's3']

--- Switch: s1 (32 rules) -----

#	Pri	Pkts	Bytes	Dur(s)	Match	Actions	Status
1	1	412	17304	1974.4	in_port=4, src_mac=00:00:00:00..drop		ACTIVE
2	1	327	13734	1974.4	in_port=3, src_mac=00:00:00:00..drop		ACTIVE
3	1	638590481268208002021974.3			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
4	1	10995362046180520401753.7			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
5	1	5603662323535381661732.7			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
6	1	10408814943717022581720.1			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
7	1	5934814724926221741698.7			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
8	1	14031118589306956 1646.3			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
9	1	7821558 328505436 1584.3			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
10	1	5739510724105944941477.8			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
11	1	23532125988349250 1123.6			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
12	1	7668809 322089978 1116.9			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
13	1	22060112926524704 1035.4			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
14	1	3461843 145397406 1007.9			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
15	1	14808723621966366 897.3			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
16	1	11276532473614344 863.7			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
17	1	11612368487719456 857.8			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
18	1	10932659459171678 812.0			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
19	1	946057 39734394 394.1			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
20	1	2746513 115353546 365.2			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
21	1	1049872 44094624 300.0			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
22	1	2158603 90661326 274.0			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
23	1	784908 32966136 169.7			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
24	1	0 0 8.7			in_port=3, src_mac=00:00:00:00..output:4		NEW
25	1	0 0 3.8			in_port=4, src_mac=00:00:00:00..output:3		NEW
26	1	0 0 3.8			in_port=4, src_mac=00:00:00:00..output:3		NEW
27	1	0 0 0.1			in_port=2, src_mac=00:00:00:00..output:3		NEW
28	1	0 0 0.1			in_port=2, src_mac=00:00:00:00..output:3		NEW
29	1	3912383 383413534 1741.0			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
30	1	2714363 266007574 1630.1			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
31	1	2645371 259246358 1594.5			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
32	0	36637 3328953 1977.8			ANY (table-miss) CONTROLLER:128		DEFAULT

--- Switch: s2 (31 rules) -----

#	Pri	Pkts	Bytes	Dur(s)	Match	Actions	Status
1	1	329	13818	1992.9	in_port=2, src_mac=00:00:00:00..output:3		ACTIVE
2	1	331	13902	1992.9	in_port=1, src_mac=00:00:00:00..output:4		ACTIVE
3	1	642423709269817957781992.7			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
4	1	11069506846491928561839.6			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
5	1	5994384025176412801750.7			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
6	1	10508751844136757561705.5			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
7	1	14165277594941634 1695.3			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
8	1	7858916 330074472 1663.8			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
9	1	5623821723620051141604.2			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
10	1	5818684124438473221462.3			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
11	1	7818511 328377462 1158.1			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE
12	1	2411996810130386561085.3			in_port=3, src_mac=00:00:00:00..output:4		ACTIVE
13	1	3508604 147361368 1054.3			in_port=4, src_mac=00:00:00:00..output:3		ACTIVE

Figure 3a: Flow table snapshot — s1 (32 rules), s2 (31 rules)

SUBJECT CODE: UE24CS252
STUDENT NAME: CHIRANTH D NANDI
SRN: PES2UG24AM048

```

Flow #1
Switch:      s1
Priority:     1
Table:       0
Cookie:      0x0
Match:       in_port=4, src_mac=00:00:00:00:00:06, dst_mac=00:00:00:00:00:05
Actions:     drop
Packets:     412
Bytes:       17304
Duration(s): 1974.438
Idle Timeout: 60
Hard Timeout: none
Status:      ACTIVE

Flow #2
Switch:      s1
Priority:     1
Table:       0
Cookie:      0x0
Match:       in_port=3, src_mac=00:00:00:00:00:04, dst_mac=00:00:00:00:00:02
Actions:     drop
Packets:     327
Bytes:       13734
Duration(s): 1974.389
Idle Timeout: 60
Hard Timeout: none
Status:      ACTIVE

Flow #3
Switch:      s1
Priority:     1
Table:       0
Cookie:      0x0
Match:       in_port=4, src_mac=00:00:00:00:00:03, dst_mac=00:00:00:00:00:01
Actions:     output:3
Packets:     638590481
Bytes:       26820800202
Duration(s): 1974.324
Idle Timeout: 60
Hard Timeout: none
Status:      ACTIVE

```

Figure 3b: Detailed rule info and aggregate summary

Mode 3 — Raw Dump

[illegible]

Figure 4: Raw OVS dump — full flow fields including ARP, ICMP, VLAN

6. Conclusion

The Multi-Switch Flow Table Analyzer meets all objectives: real-time flow retrieval across three switches, rule classification, and both snapshot and live-monitoring modes.